



Ain Shams University

Department of Mechatronics Engineering

MCT333s (UG2018) – Mechatronics System Design

Date: [13/5/2025]

Submitted by: [Team 10]

Submitted to:

Dr. Mohamed Nabil

Dr. Nancy Emad

Eng. Hossam Moataz Elkeshky

Eng. Mazen Talaat

Name	ID	Sec.
Adham Sabry	2100703	2
Ahmed Ismail	2100722	2
Kholoud Ahmed	2100712	3
Taha Gamal	2100566	2
Marwa Yasser	2101423	2
Nour Emad	2101443	2

1. Abstract

This report documents the design, simulation, implementation, and performance evaluation of a two-degree-of-freedom (2 DOF) motion control system for a robotic arm inspection platform. The system includes two independently controlled DC motors: one to rotate the base of the robotic arm and another to rotate a product platform bearing ArUco markers. PID controllers were developed and tuned for each subsystem using MATLAB/Simulink and implemented on a physical prototype using an Arduino microcontroller. System performance was evaluated in terms of response time, steady-state error, and stability. The results confirm that the implemented closed-loop control system meets precision and robustness requirements, with simulation results closely matching physical system behavior.

Table of Contents

- 1. Abstract**
- 2. Introduction**
- 3. Objective**
- 4. System Description**
 - 4.1 Subsystem 1: Robot Base Rotation
 - 4.2 Subsystem 2: Product Platform Rotation
 - 4.3 V-Shape Configuration of the Robot Arm
- 5. Mechanical Design**
 - 5.1 Concept and Design Considerations
 - 5.2 Actuator Selection and Sizing
 - 5.3 CAD Model and Fabrication
- 6. Mathematical Modeling**
 - 6.1 Subsystem 1: Robot Base Rotation
 - 6.2 Subsystem 2: Product Platform Rotation

7. Control System Architecture

7.1 Control Loop Structure

7.2 Block Diagram

7.3 PID Control Strategy

8. Simulation of Closed-Loop System

9. Hardware Implementation

10. Performance Evaluation

10.1 Comparison of Tuning Sets

10.2 Final Tuning Results

11. Conclusion

12. References

13. Appendices

- Appendix A: MATLAB/Simulink Diagrams
 - Appendix B: Motor Datasheets
 - Appendix C: Drive Links
-

2. Introduction:

Modern industrial automation and robotics applications demand high precision and stability in motion control systems. The growing complexity of robotic platforms necessitates robust and reliable control strategies that can be simulated and verified before physical deployment. This report presents the design and realization of a Two Degree of Freedom (2 DOF) motion control system. The system is designed for a visual inspection robotic platform, which consists of a rotating base for the robotic arm and a second independently rotating product platform equipped with visual targets (ArUco markers). The 2 DOF system plays a crucial role in aligning the inspection system with various targets on the product in a stationary configuration.

Closed-loop feedback control using Proportional-Integral-Derivative (PID) controllers was implemented to ensure precise rotational positioning and speed regulation of both degrees of freedom. The control design was developed in MATLAB/Simulink and later deployed on a microcontroller (Arduino uno) for real-time operation. Sensor feedback was obtained using potentiometers for angular position and DC motors served as actuators for both the robotic base and product platform.

The aim of this project is to provide a complete workflow—from system modeling and simulation to hardware implementation and performance analysis. The study evaluates the system's ability to maintain accuracy and stability under various operating conditions using metrics such as rise time, settling time, overshoot, and steady-state error. This documentation will be useful for understanding how traditional control methodologies like PID can be applied in educational and research-oriented robotics projects.

3.Objectives:

The control objective in this project is to precisely rotate two DC motors: one for the robotic base and another for the product inspection platform. Each motor is controlled independently using a **closed-loop PID control system** implemented on an Arduino microcontroller. The feedback signal is obtained from analog position sensors (e.g., potentiometers) or encoders.

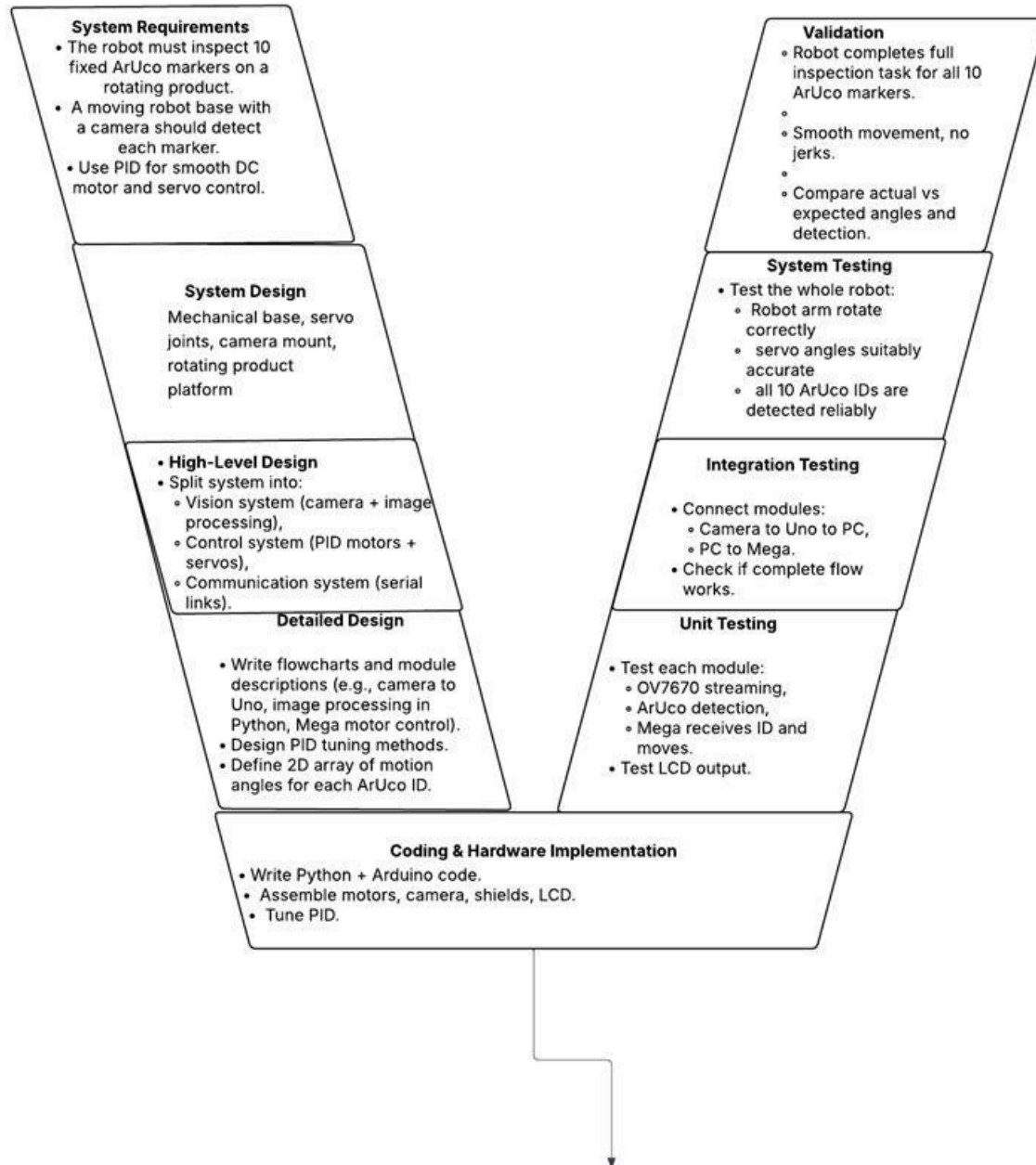
2. System Description:

The physical setup includes:

- **Subsystem 1:** A DC motor rotating the robot arm base
- **Subsystem 2:** A DC motor rotating a product platform with ArUco markers
- **Sensors:** Rotary encoders or potentiometers for position feedback
- **Actuators:** DC motors
- **Microcontroller:** Arduino uno
- **Software:** MATLAB/Simulink for simulation and deployment

Our visual inspection robot consists of a rotating base platform, a 4-DOF robotic arm with an attached camera, and a target turntable with fixed ArUco markers. The system integrates mechanical movement, computer vision, and embedded control. Image data is captured by the OV7670 camera, transmitted to a PC for processing, and then relayed to a second microcontroller that commands motion via servos and DC motors.

4.3 V SHAPE of our Robot arm system:



5. Mechanical Design:

5.1 Concept and Design Considerations

- PID Control Theory

We used Proportional–Integral–Derivative (PID) control to maintain precise rotation for both the robot's base and the turntable.

- Implementation for DC Motors

PID was implemented in Arduino using feedback from rotary encoders. Target angles were matched with actual encoder readings to compute error and drive the motors.

- Encoder Feedback Integration

Each DC motor was equipped with an encoder. PID loop runs at a fixed interval and calculates speed and position to ensure accurate stopping and minimal overshoot.

The robot consists of a rotating base (using a DC motor with PID control) and an articulated arm composed of three links plus a camera mount, each controlled by a servo motor. The design is stationary, with only the base rotating to cover a 360-degree range.

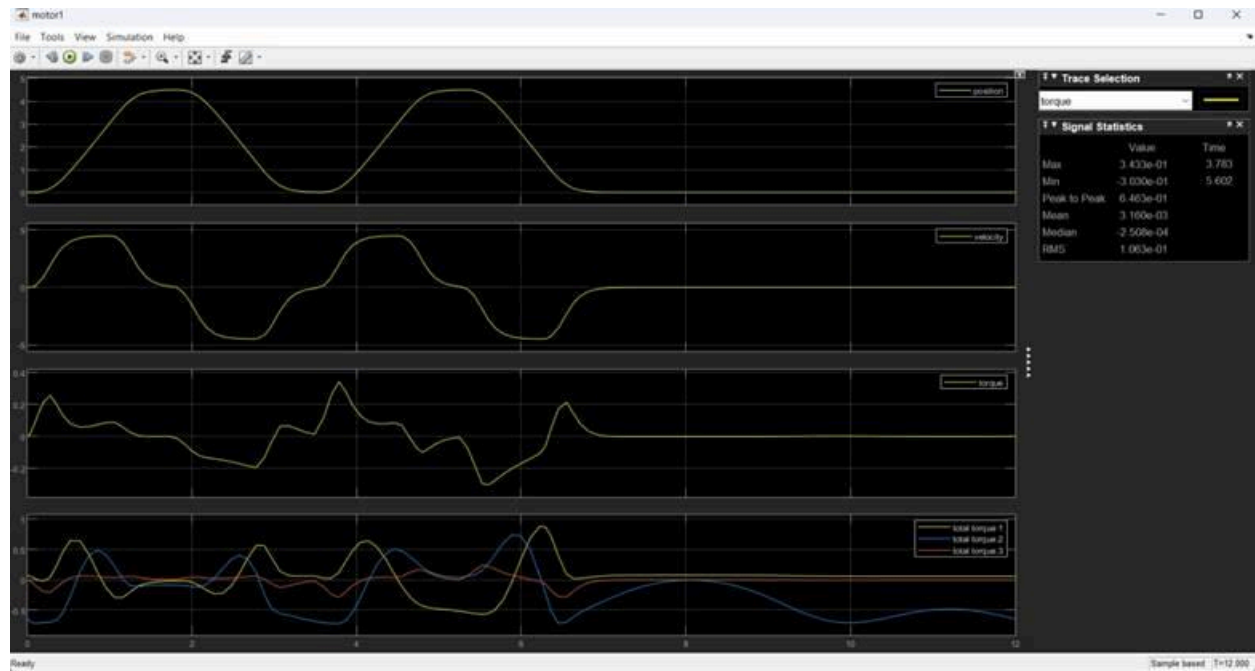
- **Rotating Base:** Allows full inspection coverage.
- **Link 1 & Link 2:** Provide vertical and horizontal reach.
- **Link 3:** Additional movement precision.
- **Camera Mount Servo:** Adjusts camera angle for optimal detection.
- **Target Platform:** The inspected product is placed on a rotating base (DC motor + PID).

5.2 Actuator Selection and Sizing

- **Base DC Motor (Robot Rotation):** Selected for high torque and compatibility with PID control.
- **DC Motor (Target Platform):** Used to rotate the turntable with ArUco targets.
- **Servo Motors:** Four servos are used: three for the robot arm joints and one to adjust camera orientation.

Actuator sizing

- MOTOR 1



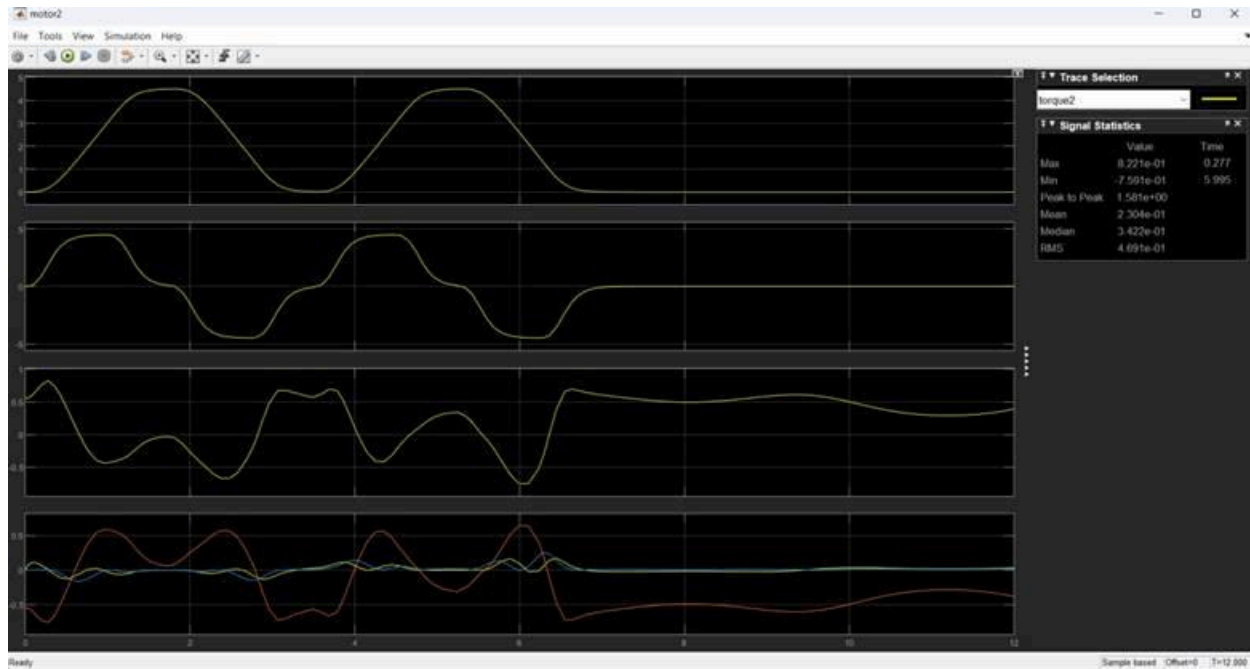
- MAX TORQUE: $0.35 \text{ N.M} \times \text{safety ratio}(1.5) = 0.525 \text{ N.m}$
- MAX VELOCITY : 4.5 rad/sec
- Available motor



So we will use these motor with gear ratio = N

Torque motor= $0.4315 \times 3 = 1.3 \text{ N.M}$

MOTOR 2



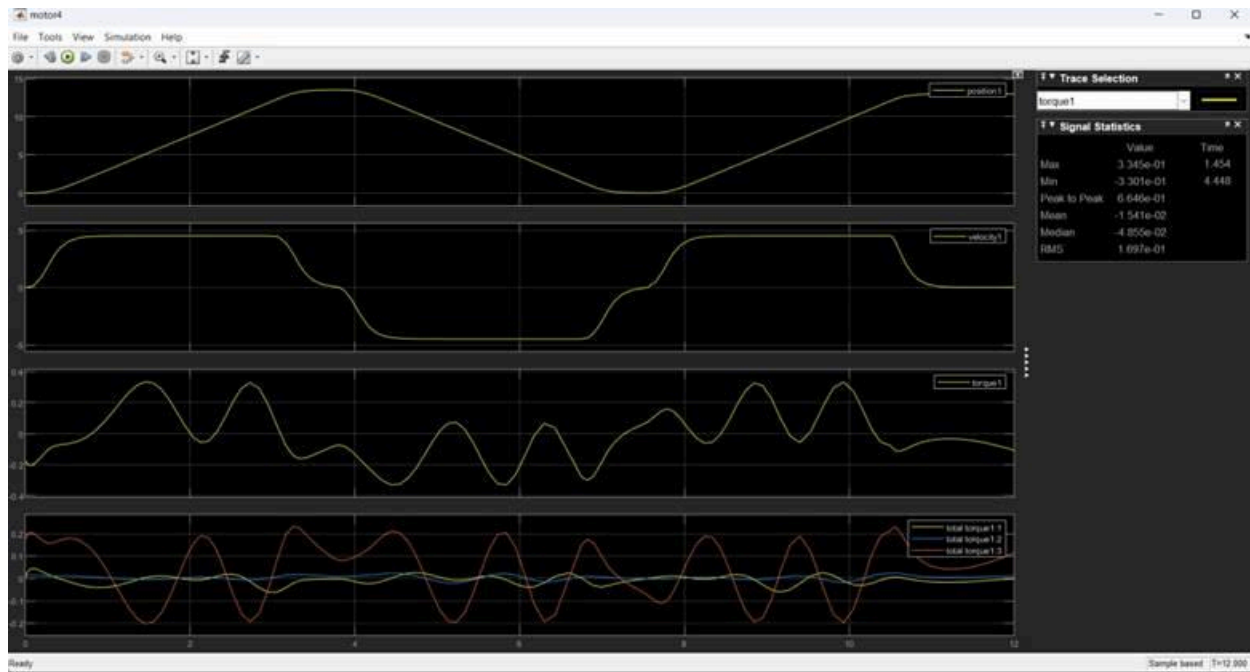
MAX TORQUE: $0.81 \text{ N.M} \times \text{safety ratio}(1.5) = 1.215 \text{ N.m}$

MAX VELOCITY : 4.5 rad/sec

❖ Available motor specification in datasheets attached in **Appendices**

Torque motor= **1.373 N·m**

MOTOR 3



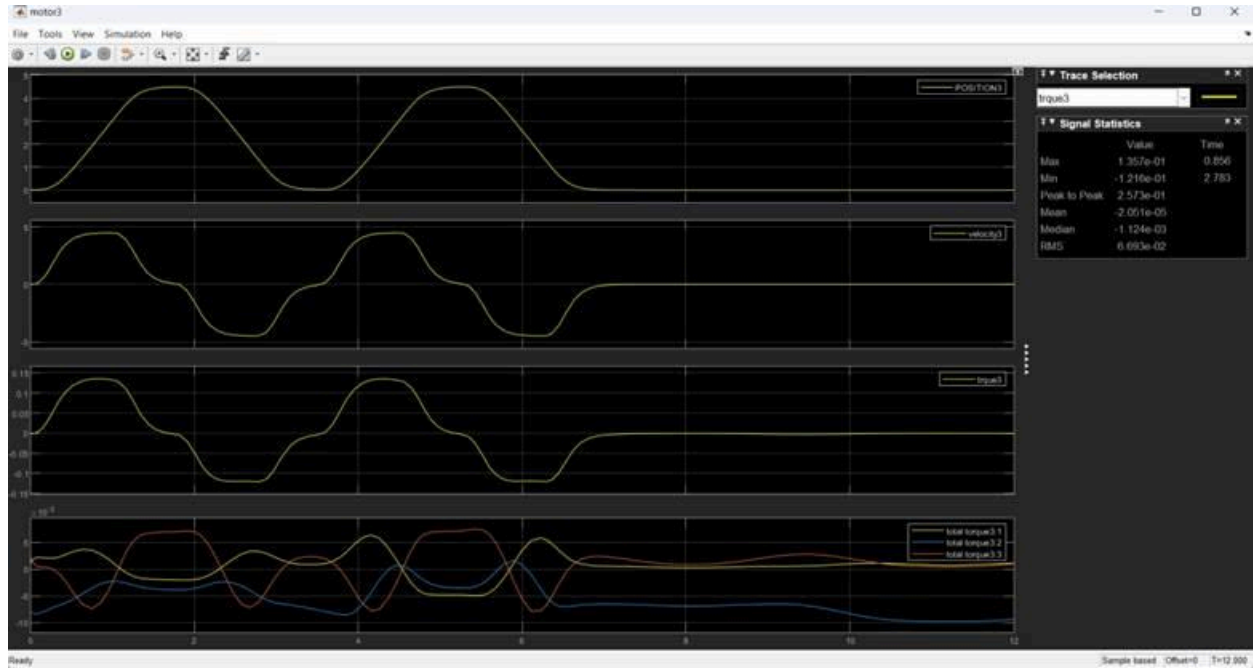
MAX TORQUE: $0.35 \text{ N}\cdot\text{m} \times \text{safety ratio}(1.5) = 0.525 \text{ N}\cdot\text{m}$

MAX VELOCITY : 4.5 rad/sec

❖ Available motor specification in datasheets attached in **Appendices**

Torque motor= **1.373 N·m**

MOTOR 3 MOTOR 4



· MAX TORQUE: $0.12 \text{ N.m} \times \text{safety ratio}(1.5) = 0.18 \text{ N.m}$

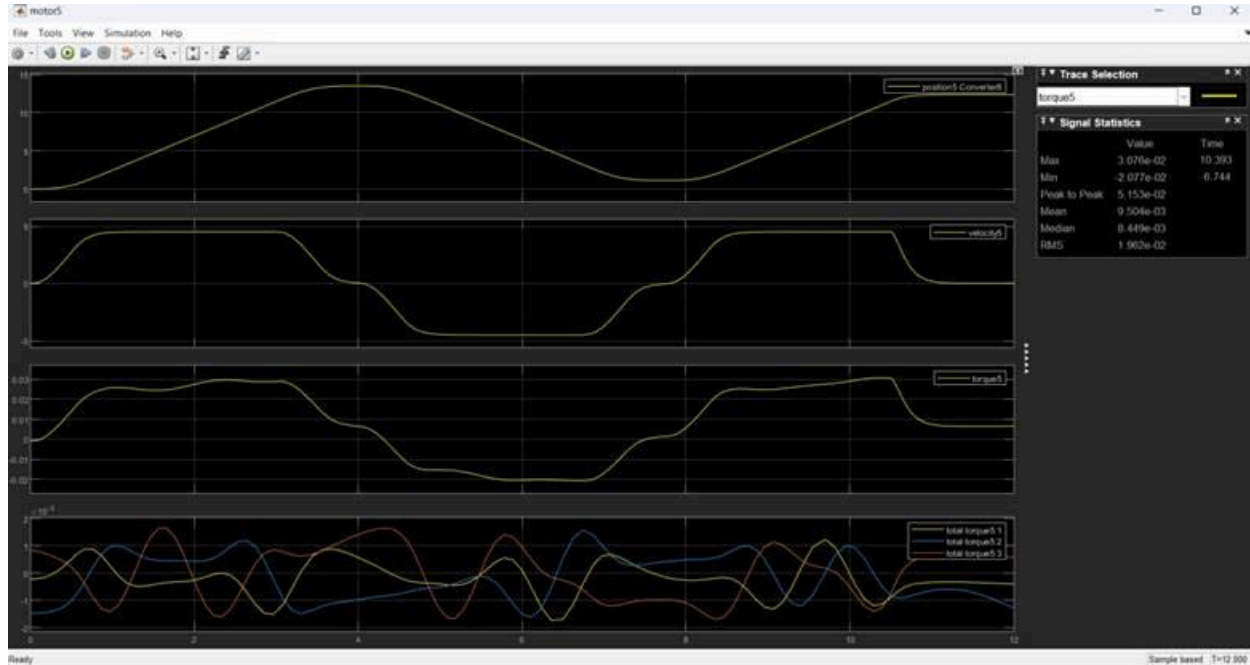
· MAX VELOCITY : 4.5 rad/sec

❖ Available motor

specification in datasheets attached in **Appendices**

Torque motor= **0.2942 N.m**

MOTOR 5



· MAX TORQUE: $0.03 \text{ N}\cdot\text{m} \times \text{safety ratio}(1.5) = 0.045 \text{ N}\cdot\text{m}$

· MAX VELOCITY : 4.5 rad/sec

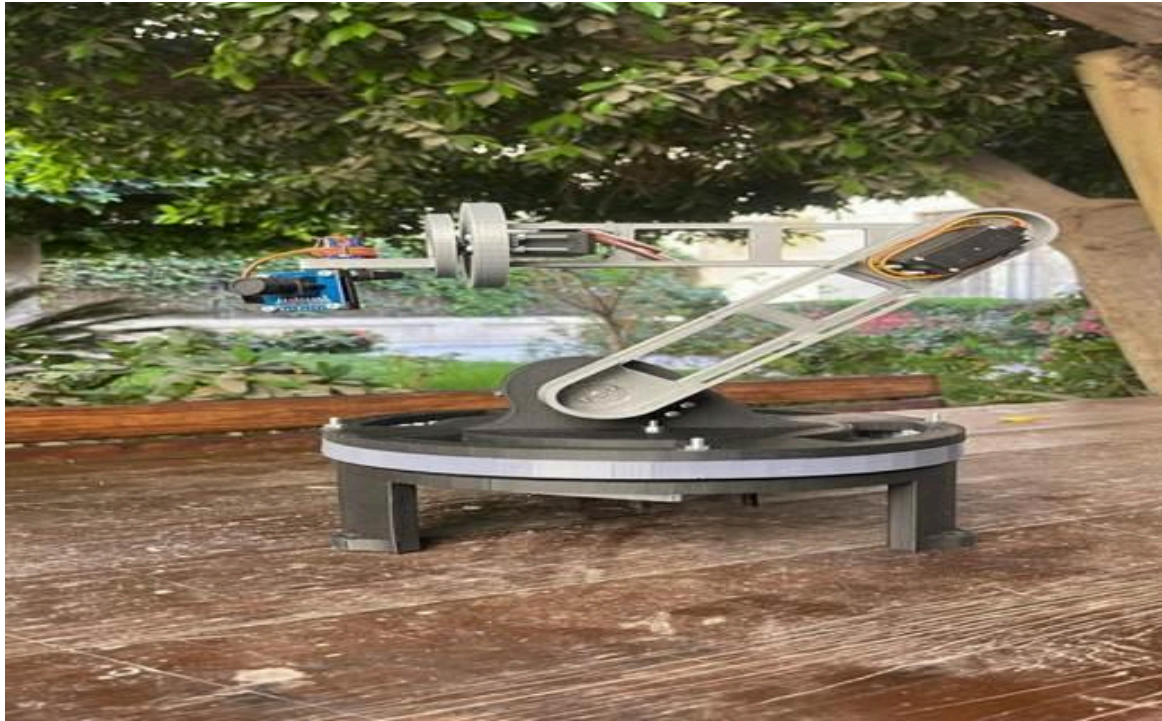
❖ Available motor specification in datasheets attached in **Appendices**

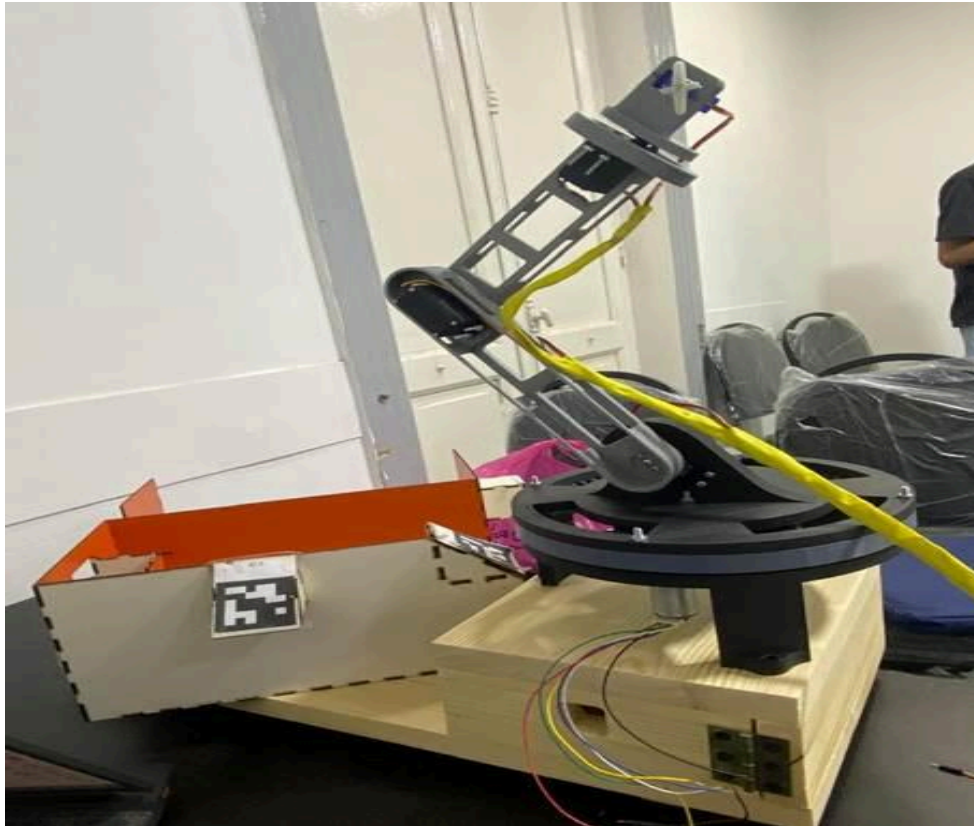
· Torque motor= **$0.1471 \text{ N}\cdot\text{m}$**

5.3 CAD Model and Fabrication

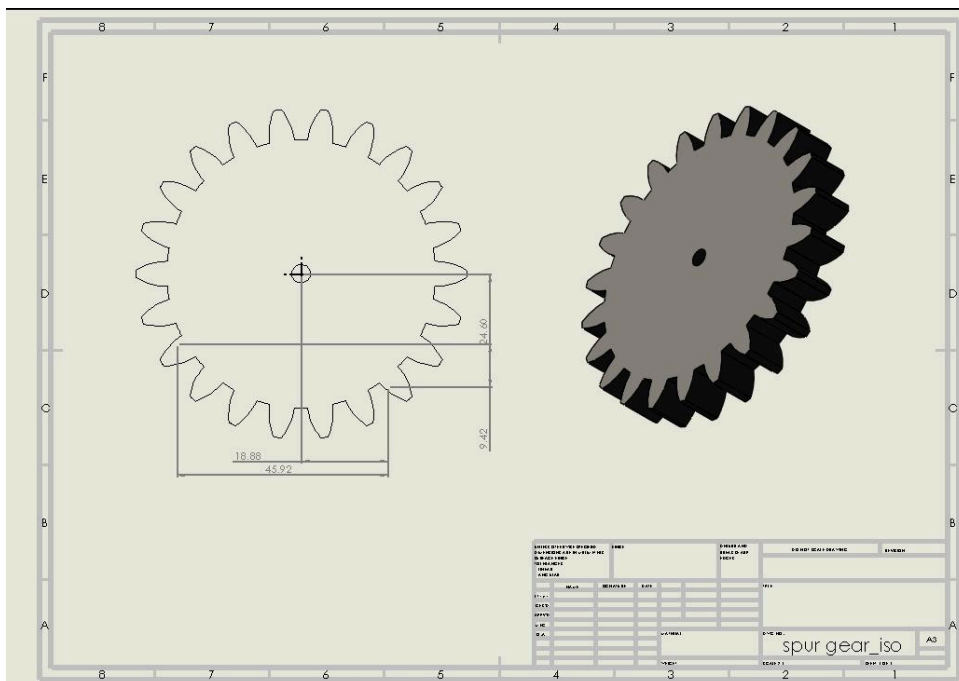
The design was created using SolidWorks. The parts were fabricated using 3D printing and laser-cut acrylic. The camera was mounted securely at the end-effector of the arm. The base and turntable were fixed onto MDF platforms for stability.

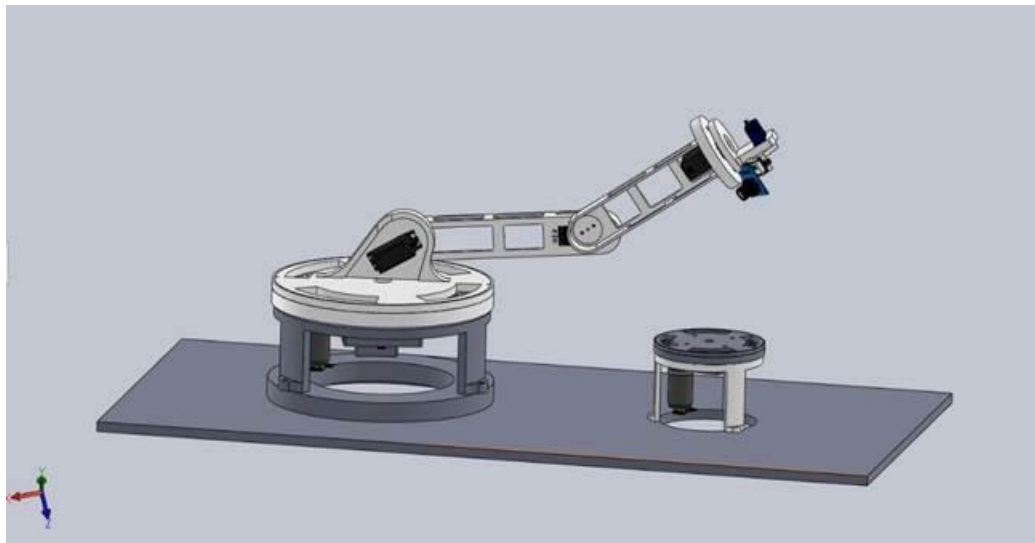
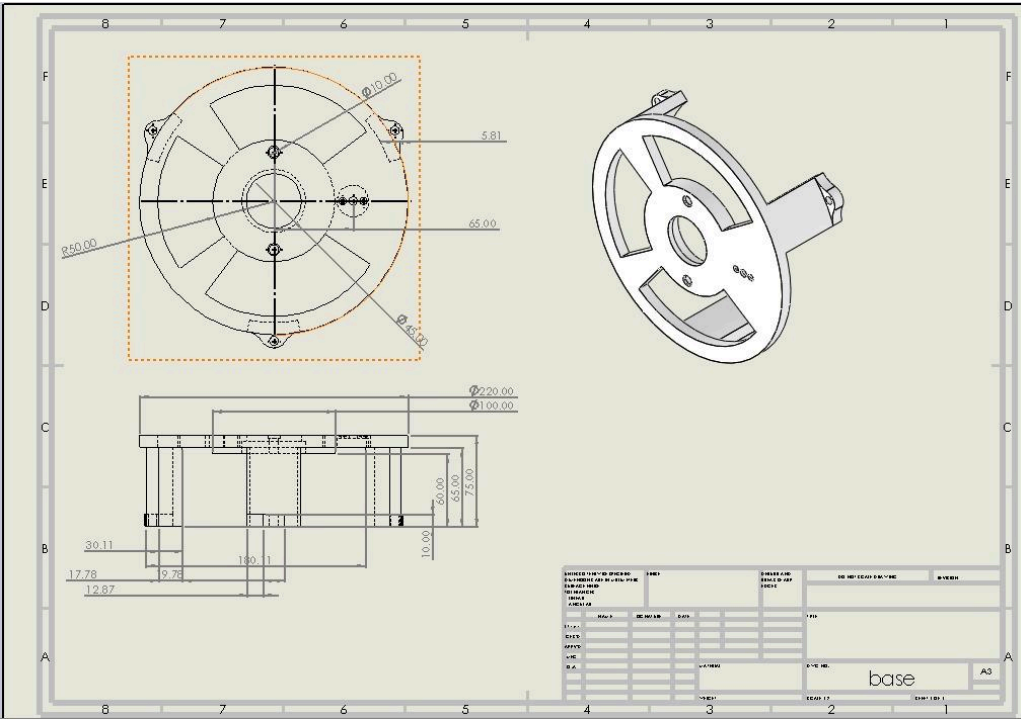
The whole system



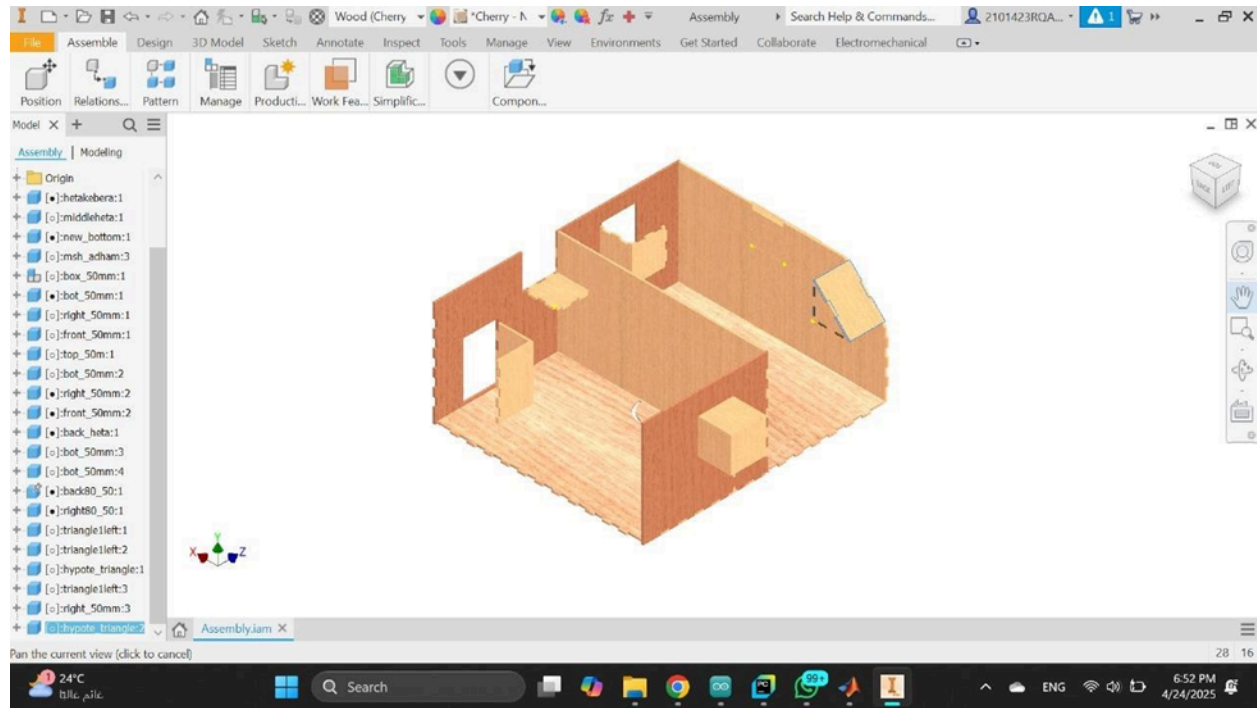


Base of robotic arm





product



6. Mathematical Modeling

6.1 Subsystem 1: Robot Base Rotation (DC Motor)

Transfer Function:

Where:

- : Moment of inertia
- : Armature inductance
- : Armature resistance
- : Viscous friction coefficient
- : Motor torque constant

- : Back EMF constant

•

1.1. Electrical Equation (from Kirchhoff's Law):

$$V(t) = L \frac{di(t)}{dt} + Ri(t) + K_e \frac{d\theta(t)}{dt}$$

1.2. Mechanical Equation (from Newton's Second Law):

$$J \frac{d^2\theta(t)}{dt^2} + B \frac{d\theta(t)}{dt} = K_t i(t)$$

التاريخ: Mathematical Model

⇒ elec eq $V(t) = \underbrace{L \frac{di(t)}{dt}}_{\text{Inductance}} + \underbrace{R i(t)}_{\text{Resis}} + \underbrace{K_e \frac{d\theta(t)}{dt}}_{\omega(t)}$

⇒ Mechanical eq $J \frac{d^2 \theta(t)}{dt^2} + B \frac{d\theta(t)}{dt} = \underbrace{K_t i(t)}_{\substack{\text{Armature} \\ \text{Current}}} \rightarrow \underbrace{T(t)}_{\text{Torque}}$

Transfer fun $G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{(Ls+R)(Js^2+Bs) + K_e K_t}$

Motor
Transfer
Const

⇒ Laplace Transform:

elec sys $V(s) = (Ls+R)I(s) + K_e s \theta(s)$

Mechan sys $J s^2 \theta(s) + B s \theta(s) = K_t I(s)$

from Mechanical equation

$$I(s) = \frac{J s^2 + B s}{K_t} \theta(s)$$

Sub in elec equation:

$$V(s) = [Ls+R] \cdot \frac{J s^2 + B s}{K_t} \theta(s) + K_e s \theta(s)$$

$$\Rightarrow V(s) = \left(\frac{(Ls+R)(Js^2+Bs)}{K_t} + K_e s \right) \theta(s)$$

$$G(s) = \frac{\theta(s)}{V(s)} = \frac{K_t}{(Ls+R)(Js^2+Bs) + K_e K_t s}$$

* Defining that

$$R = 1 \Omega$$

$$L = 0.01 H$$

$$I_0 = 0.01 A$$

$$B = 0.1 A/m \cdot s$$

$$K_t = 0.01 N \cdot m/A$$

$$K_v = 0.01 V \cdot s/A$$

Transfer function

$$\Rightarrow \frac{\Theta(s)}{V(s)} = \frac{0.01}{0.01 s^2 + 0.01 s + 0.001}$$

6.2 Subsystem 2: Product Platform Rotation

Same structure as Subsystem 1 with respective parameters.

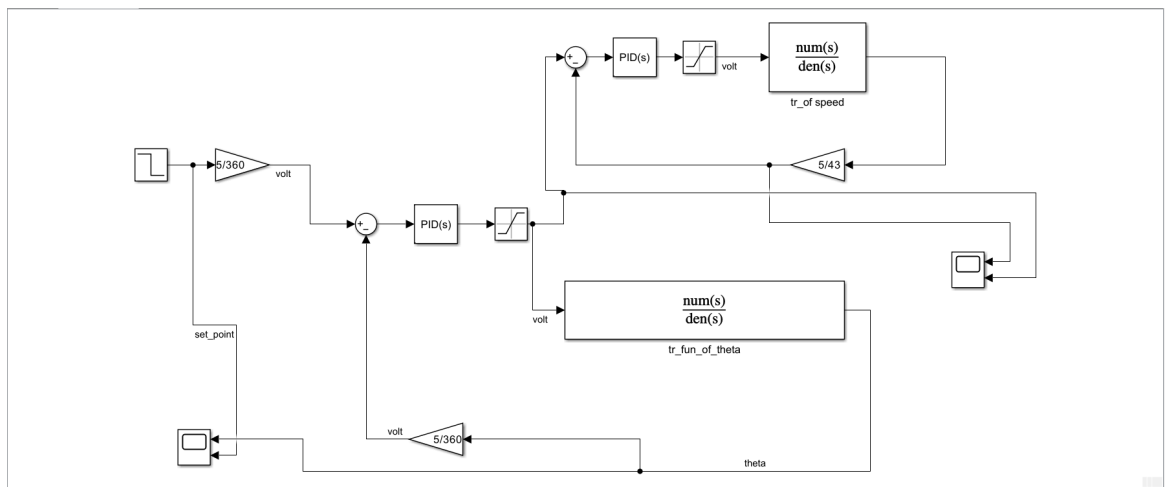
7. Control System Architecture

7.1 Control Loop Structure

Each subsystem uses a **single-loop PID controller for position control**. If using encoders with high enough sampling speed, an **inner loop** for speed control could be implemented; however, in this project, the focus is on accurate **position control**, so only the outer position loop is considered. The **typical PID loop structure** used

7.2 Block Diagram

- **simulink blocks**



- **Transfer function of angle**

$$\frac{1}{(L * jeq)s^3 + ((L * Deq) + (Rm * jeq))s^2 + ((Rm * Deq) + (Ke * Kt))s}$$

tr_fun_of_theta

- Transfer function of speed

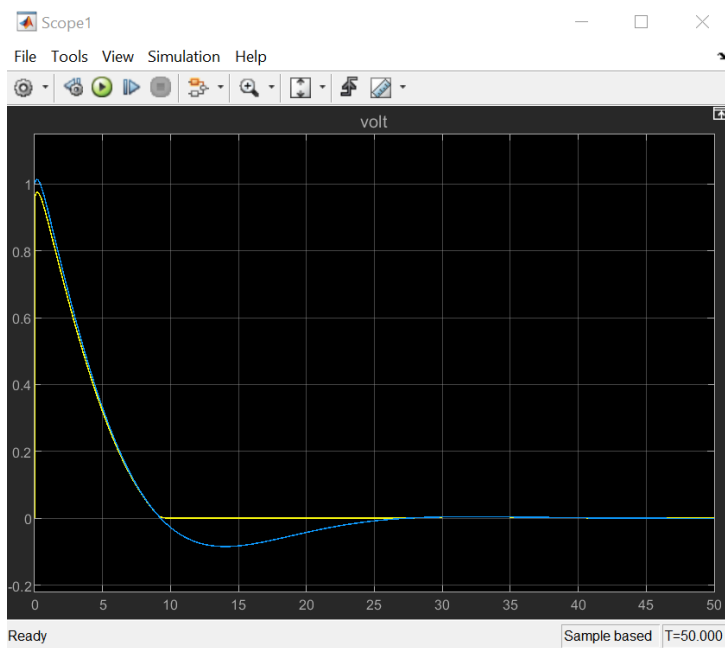
$$\frac{1}{(jeq * Rm)s + ((Ke * Kt) + (Rm * Deq))}$$

tr_of speed

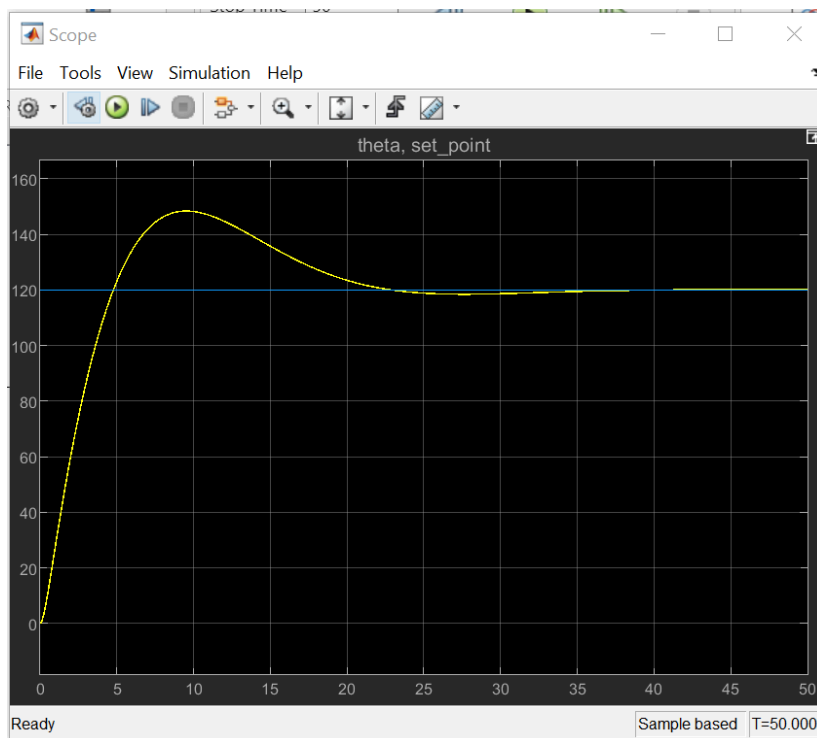
- system parameters

```
>> K1=1;
>> L=0.0078094;
>> Deq=0.0026866;
>> jeq=0.00065033;
>> Rm=10;
>> Kt=0.01;
>> Ke=0.01;
```

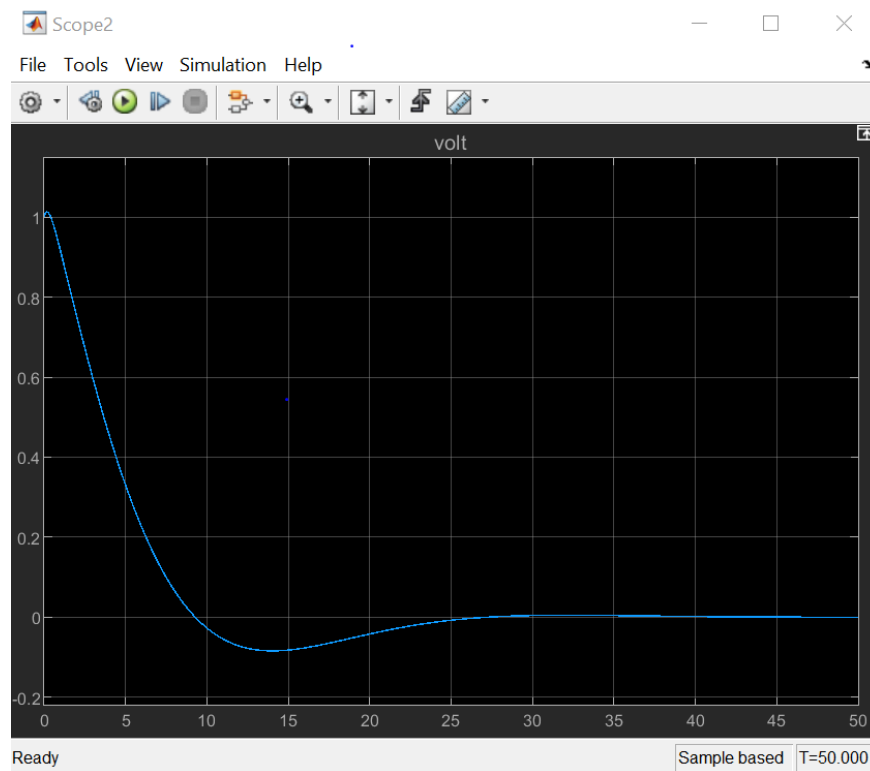
- system response (speed)



- **system response (angle)**



- **control action of first pid (position):**



7.3 PID Control Strategy

Each subsystem is controlled via a closed-loop PID controller implemented on the Arduino. The strategy includes:

- Position Control: Outer PID loop
- Speed Control: Optional inner loop (not used here)

Tuning Methods:

- Simulink PID Tuner: Used to generate ideal gains for simulation
- Manual Tuning: Applied after observing real-world system behavior
- Ziegler-Nichols Method: Used to cross-check and validate gain ranges

8. Simulation of Closed-Loop System

Simulink models included:

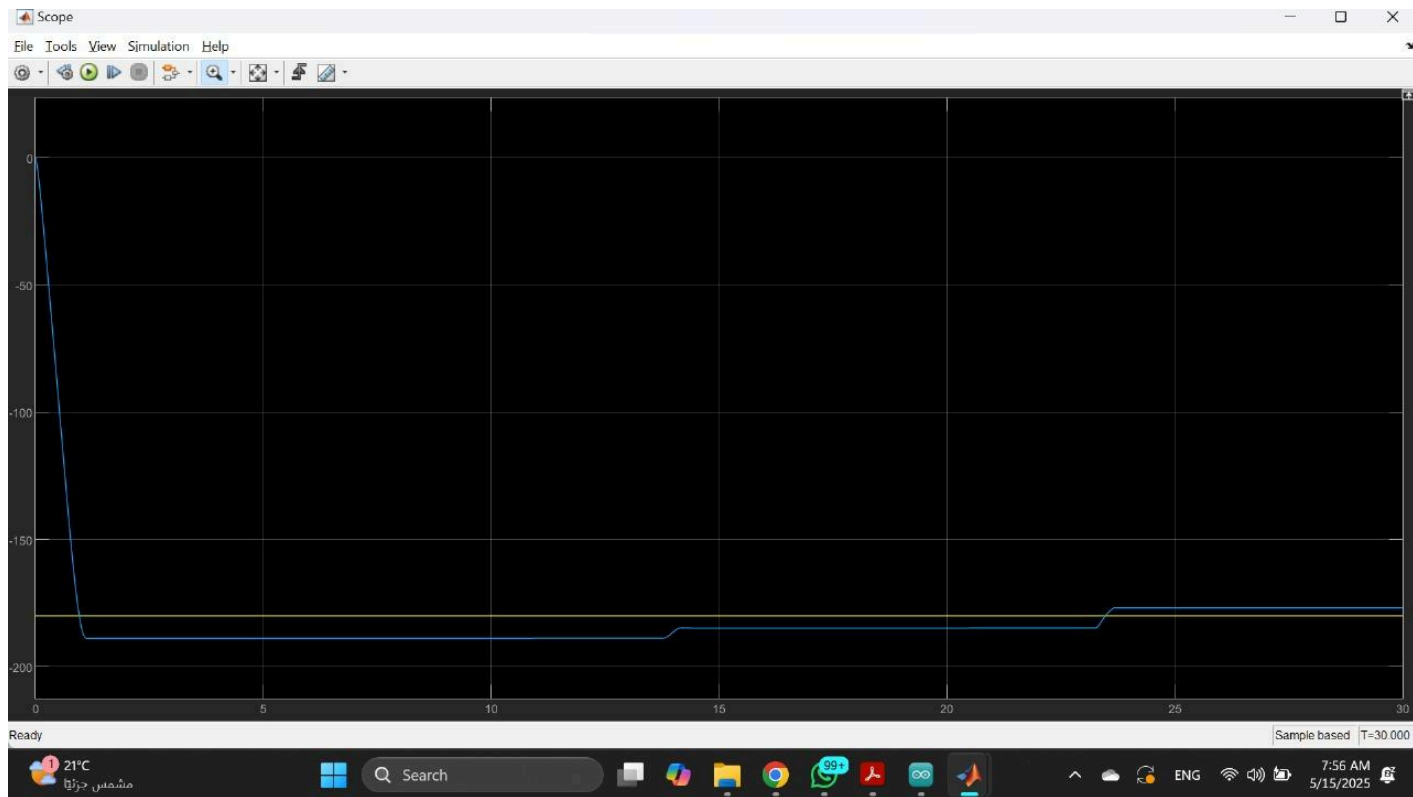
- DC motor plant (transfer function model)
- PID Controller block
- Scope blocks to observe
- Step input and disturbance signals
- Developed in Simulink using Transfer Function and PID blocks
- Used step input and disturbance rejection scenarios
- Captured response curves: $r(t)$, $y(t)$, and $u(t)$
- Scope blocks used for signal monitoring

Simulation goals:

- Minimize overshoot and settling time
- Ensure system stability and responsiveness

9. Hardware Implementation

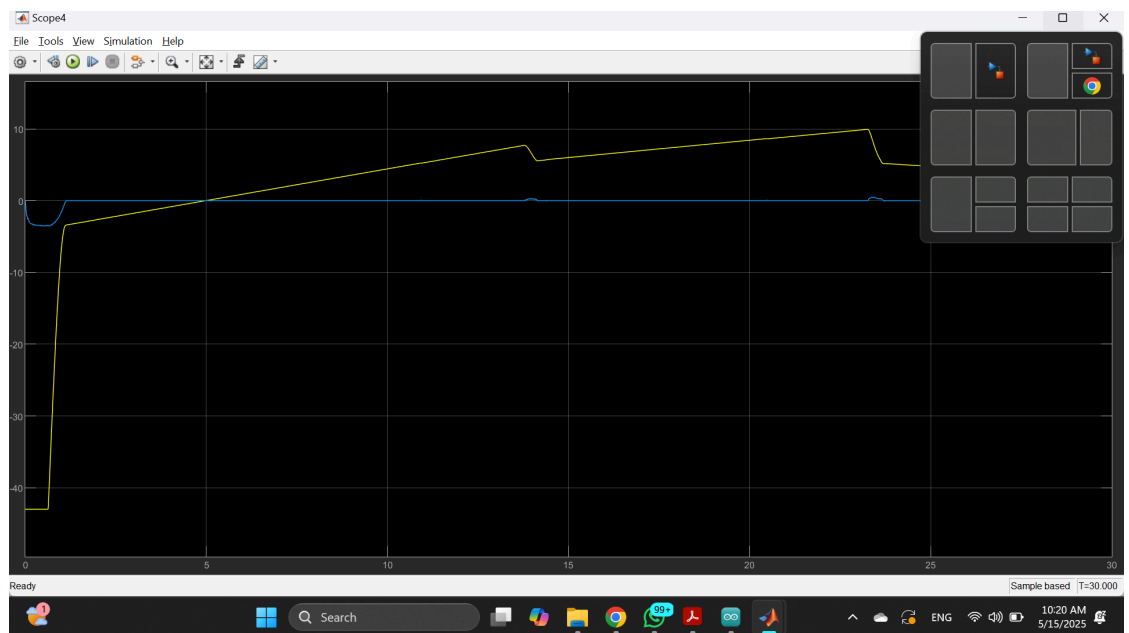
- Controller implemented on Arduino E using Simulink support package
- Real-time control signals measured via serial monitor
- Signals displayed:
 - Setpoint ($r(t)$) & Measured angle ($y(t)$)
 - ❖ of the arm: with set point = -180
 - $k_p = 0.6$
 - $k_i = 0.1$
 - $k_p = 6$ (second loop)



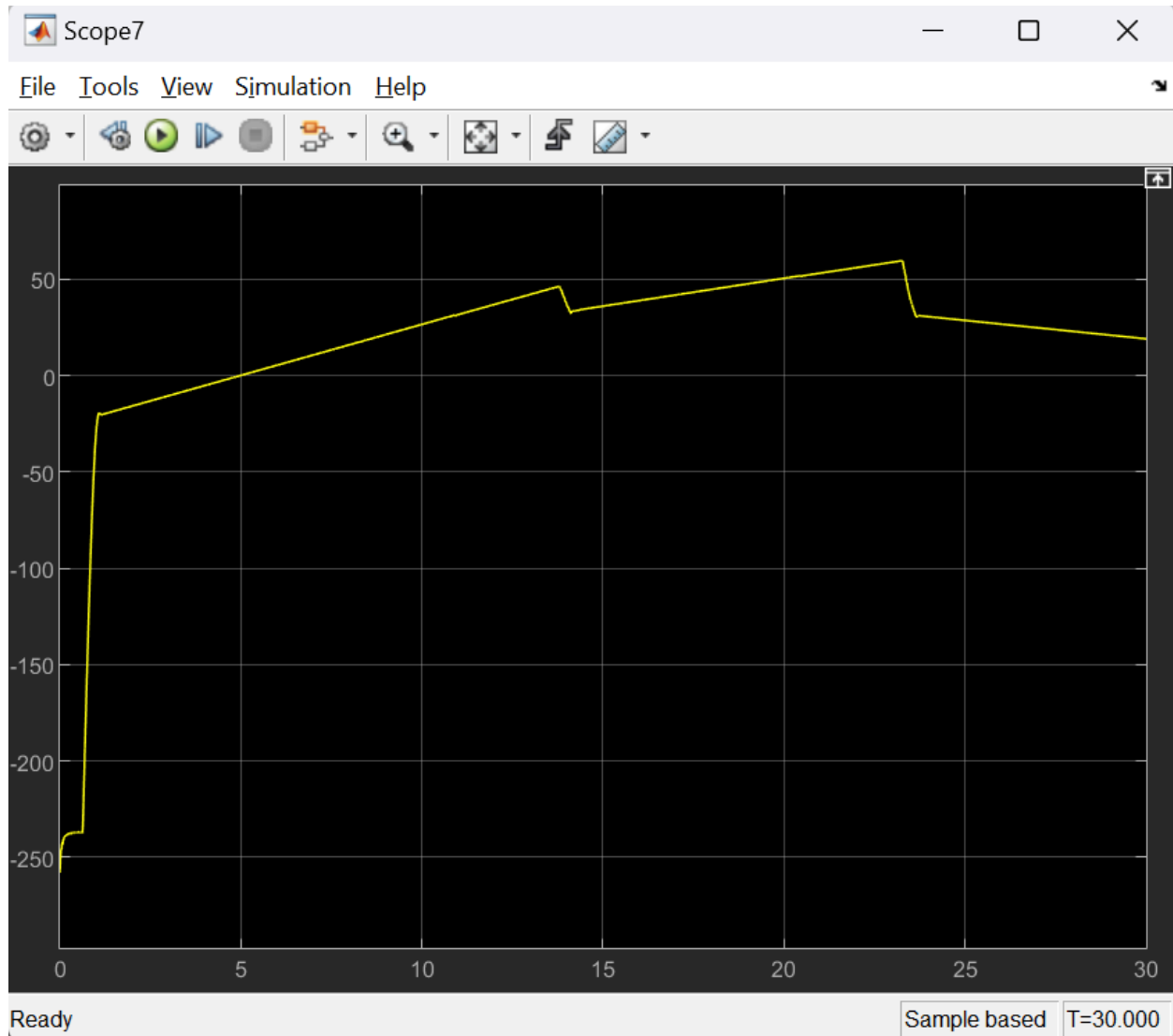
- ❖ of the base:
 - with set point = **180**
 - $k_p = 0.6$
 - $k_i = 0.1$ for pos
 - $k_p = 6$ (second loop)



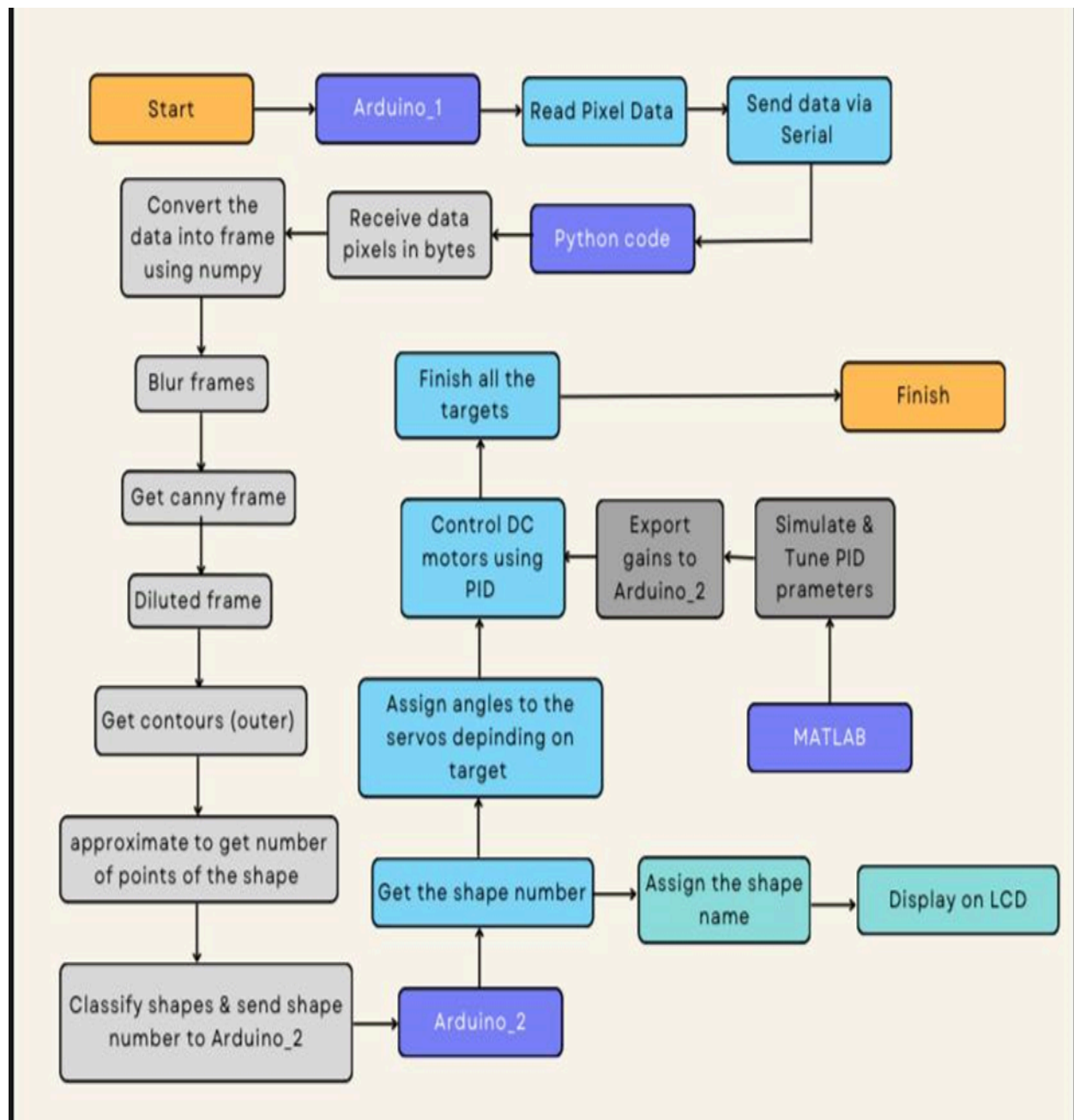
-
- Control signal ($u(t)$)
- Control signal ($u(t)$) of the arm for the first loop



control action of the arm of the second loop:



Flow charts how the whole system is integrated together



10. Performance Evaluation

10.1 Comparison of Tuning Sets

Tuning Set	Rise Time	Settling Time	Overshoot	Steady-State Error
------------	-----------	---------------	-----------	--------------------

10.2 Final Tuning Results

- Stable response
- High accuracy

11. Conclusion

The 2 DOF motion control system was successfully modeled, simulated, and implemented. PID controllers provided accurate and stable positioning for both the robotic base and the rotating platform. The simulation results closely matched experimental results, validating the control design. Future work could involve adaptive PID tuning or integrating machine vision feedback into the control loop.

12. References

1. Ogata, K. (2010). Modern Control Engineering. Prentice Hall.
2. MATLAB & Simulink Documentation – MathWorks
3. Nise, N. S. (2011). Control Systems Engineering. Wiley.

13. Appendices

Appendix A:

MATLAB/Simulink Diagrams

(Insert screenshots of the simulation model)

Datasheets

motor 2

Specifications

Voltage	6 -7.2V
Speed	0.16 sec/60 degree (6V), 0.14 sec /60 degree (7.2v)
Torque	14 kg.cm (6V), 15 kg.cm (7.2V), Stall torque
Dimension	40.8 x20.1x38 mm
Weight	56g
Horn Kit	Set of 4 pieces of arm servo horn with different sizes and shapes - 2 horn mounting screws, 4 servo mounting screws, 4 rubber vibration absorbers for servo mounting

motor 3

Specifications

Voltage	6 -7.2V
Speed	0.16 sec/60 degree (6V), 0.14 sec /60 degree (7.2v)
Torque	14 kg.cm (6V), 15 kg.cm (7.2V), Stall torque
Dimension	40.8 x20.1x38 mm
Weight	56g
Horn Kit	Set of 4 pieces of arm servo horn with different sizes and shapes - 2 horn mounting screws, 4 servo mounting screws, 4 rubber vibration absorbers for servo mounting

motor 4

Specifications

Voltage	4.2-6V
Speed	0.18 sec/60degree (4.8v), 0.16 sec/60degree (6v)
Torque	3 kg.cm (4.8v), 3.2 kg.cm (6v)
Dimension	40.8 x 20.1x 38 mm
Weight	36g
Horn Kit	Set of 4 pieces of arm servo horn with different sizes and shapes - 2 horn mounting screws, 4 servo mounting screws, 4 rubber vibration absorbers for servo mounting

motor 5

Features

- Continuous rotation
- Operating voltage: 4.8-6.0VDC
- Top operating speed: 60-70RPM (4.8-6.0VDC respectively)
- Torque: 1.2-1.5 kg-cm (4.8-6.0VDC respectively)

Drive link

https://drive.google.com/drive/folders/1Qo6fTLr5qPiKK-dOJmVSESqMmpqS9kih?usp=s_haring